



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 4 REPORT

Internship Program — Batch 2026

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Date:	07 July 2026

1. Introduction

This report presents the work completed for **Task 04** of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. The internship focuses on practical exposure to core concepts in Artificial Intelligence and Machine Learning, equipping interns with skills applicable to real-world technology and marketing challenges.

M-Tech Production & Marketing is a forward-looking company based in Haripur, KPK, specializing in Technology, Training, Marketing, and Projects. The internship is structured in a 9-Phase roadmap designed to progressively build theoretical knowledge and hands-on project experience.

2. Objectives

- To understand the fundamentals of Machine Learning.
 - To learn the complete Machine Learning workflow.
 - To study the different stages involved in building a machine learning model.
 - To understand the importance of data preparation, model training, testing, and evaluation.
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3. Task Description

The assigned task was to study the core concepts of Machine Learning and understand the complete workflow followed in real-world machine learning projects. This included learning how data is collected, prepared, used for model training, evaluated, and finally deployed for predictions.

4. Work Done

During this experiment, I completed the following activities:

- Studied the fundamentals of Machine Learning and its real-world applications.
- Learned the difference between training data and testing data.
- Explored the complete Machine Learning workflow from data collection to model deployment.
- Understood the importance of data preprocessing, including data cleaning and feature preparation.
- Learned how machine learning models are trained using training datasets.
- Studied model evaluation techniques to measure prediction performance.
- Understood the prediction stage where trained models are used on new unseen data.
- Reviewed the overall lifecycle of a machine learning project and how each stage contributes to building an effective AI solution.

5. Challenges Faced

- Understanding the sequence of all workflow stages initially required careful study.
- Differentiating between model training, testing, and evaluation was slightly challenging.
- Learning the purpose of each workflow step required revising the concepts multiple times.

6. Key Learnings

- Learned the complete Machine Learning workflow from start to finish.
- Understood the role of data preprocessing in improving model performance.
- Gained knowledge of how machine learning models are trained and evaluated.
- Learned the importance of testing models before deployment.
- Improved my understanding of the overall Machine Learning development process.

7. Conclusion

This experiment provided a solid understanding of Machine Learning fundamentals and the complete workflow used in developing AI models. By studying each stage of the process, I gained a clear understanding of how machine learning projects are built, evaluated, and deployed. This experiment has strengthened my foundation for upcoming practical machine learning tasks in the internship.

Intern Signature: Hamza Badshah

Supervisor: _____

Name: Hamza Badshah

Name:

Date: 07 July 2026

Designation: